

Advanced Genomics projects – A.Y. 2025/2026

1. Functional analysis of single cell transcriptomics data. Take some single cell data where cells were grown in different conditions (see bottom). A dimensionality reduction algorithm like UMAP or t-SNE will show a clear separation of cells from different conditions. This means that the different media are inducing specific reconfiguration of the transcriptome and therefore of the metabolic fluxes to enable the utilization of different carbon sources, the synthesis of essential compounds not present in the media etc etc. One possible task of the exam project is to define what are the **functions** that determines the separation of the cells in groups specific of each medium. This can be done by analyzing the loadings of the transcripts in the PCA output and then assigning them to their respective functional categories, or better, you can try the opposite: aggregate expression level into a measure of functional activity before the PCA, and then analyse the loadings associated to the functional categories in the direction of maximum separation. It is however possible that PCA does not give a clear-cut separation of the cells as the ones you get with UMAP, at least in 2D or at most 3D plots – this means that even if you are not able to see a separation of the different data points, you may still be able, by using more than 3 components, to partition the observations by medium. You can achieve this for instance by clustering all the components such that they account for a large fraction of the variance in the data. Otherwise, you can stick on the UMAP dimensionality reduction; in this case the problem is that now you do not have something similar to the loadings of the PCA and therefore you have to perform additional analysis to understand what are the original variables determining the separation of the data points in the UMAP coordinate space. One possible way is to correlate linearly, e.g. pearson's or non-linearly e.g. using spearman's (only monotonic correlations) or mutual information (ok with monotonic and non-monotonic relationships), the UMAP coordinates with the original variables e.g. your "pathway expression" index. Since the data have a large number of observations, I suggest to work with a subsample of cells; this enables also to equalize the number of cells per medium.

Functional classifications:

<https://current.geneontology.org/products/pages/downloads.html>

or

<https://www.genome.jp/brite/sce00001>

and then downloading in htext format which encodes all the relationships gene name / metabolic "pathway". This requires some effort to be loaded properly in R, but the presence of a first character in each row indicates the hierarchy of the information contained in each section. For instance all lines after "C 00010 Glycolysis / Gluconeogenesis [PATH:sce00010]" belongs to Glycolysis.

Other resources are also admissible, if they provide the information required (functional processes/pathways).

possible transcriptomics data sources for yeast (you can do the project also on the E. coli transcriptomics data linked for other projects below):

<https://www.nature.com/articles/s42003-021-02320-w>

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165686>

GSE165686_yeastdropseq_dge.mtx.gz contains the data, the tsv files additional information you may need

MORE MEDIA TESTED:

<https://elifesciences.org/articles/51254>

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE125162>

file to download: GSE125162_RAW.tar

2. **Classification of single cell transcriptomics data.** A task related to the above is to derive a “functional signature” of the different media where cells were grown based on the functional annotation of the genome, like before, you can implement some sort of “pathway expression index” by grouping genes of the same functional category and merging their expression level (what to use? mean? Median? Check the distribution of the original expression data to understand if you need to log-transform the read counts before calculating the statistics); then you set up a classification algorithm to classify unobserved cells to the different media. Here you have to focus more on technical details about the performances of the classification algorithm, like F1 score, precision, recall etc etc to characterize in deep how the classification is going and maybe show how different data transformations or options for the classification algorithm affect the outcome. Moreover, choose a classification algorithm that enable to extract the importance of the different variables in the classification itself to be able to shortly discuss why certain pathways are more important for differentiating some media with respect to others.

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3. Cell cycle distribution from single cell transcriptomics data. Find a table indicating genes that can be used as markers of different cell cycle phases – check if you can also get some marker genes of no growth (can help in validation). Typically, these genes have cell cycle dependent expression patterns and have a peak of expression in well-defined phases, such that identifying their maximal expression enable to assign a cell cycle phase to single cells. Think about a way to combine the gene expression level of markers for the same cell cycle phase, as a first guess you could try the average, or the median of all genes of the same phase to get an index for that phase. Think about the fact that maybe the sequencing depth is not very high and some genes may have zero expression because they are not highly expressed. How to deal with these cases? One option is to remove genes with 0 expression, or to calculate the statistics only on a subset of the genes, for instance those that are more expressed. Then take a decision about how you can use you cell cycle phase indices to assign cells to phases. Cell get the phase for which you get the highest index? Or maybe you can be more stringent and assign a phase only when you are sure (for instance because the second best index is much smaller? How much smaller is right, however?).

How can you validate your assignment? For instance, if the UMAP dimensionality reduction contains information about the periodic expression of the cell cycle dependent genes, you expect cells in the same cell cycle phase to be somehow clustered in the UMAP, and you can test for this – visually, or much better quantitatively (for instance you could compare the average distance of cells grown in the same medium and assigned to the same cell cycle phase in the UMAP to the distance separating cells from the same medium but randomly sampled. You can also check the media for those where you expect cells to be in stationary phase, and given your assignment you should observe a much smaller number of represented cell cycle phases (only one at one extreme). One further validation could be the following, which is however extrapolated from growth conditions and not available in the paper metadata. Referring to the elife paper linked below, one group of cells was grown in YPD at 30°; as a first approximation, in this condition:

Total cell cycle: **~90 minutes** (consistent with $\mu \approx 0.45\text{--}0.5\text{ h}^{-1}$)

Phase	Approx. duration	Fraction of cycle
G1	25–35 min	~30–40%
S	20–25 min	~25–30%
G2	10–15 min	~10–15%
M	15–20 min	~15–20%

This means that you should get a percentage of cells in each phase similar to the percentage in the above table for the different phases. Do not worry if this is not the case, the growth rate may depend on other factors.

Once you demonstrate that somehow your assignment works, you can answer the final question of the project: are the different media very different concerning the fraction of cells in each cell cycle phase?

Can you elaborate a little bit around this? For instance, in poorer media, what cell cycle phase become more represented and why, in your opinion?

4. Single cells, transcription factor activity proxy, and post-translational modifications. The activity of a transcription factor is in principle detectable by observing the expression level of its targets. Indeed, we usually assume that genes expressed similarly are often controlled by a same regulator, and therefore also the reverse should be true. In this project, we want to introduce a proxy for a transcription factor activity based on the expression level of genes regulated by that transcription factor. The regulator-target relationships can be found in specialized databases, such as regulondb for coli, yeasttract for yeast... For a positive regulator, its activity can be taken as being a function (linear? sigmoidal?) of the expression level of its targets (mean? median? max?). Now, the question is to show the degree at which transcription factor activities are correlated to the corresponding transcription factor gene expression level. Our null hypothesis is that indeed, the abundance of the transcript is also a proxy of the activity. This can be somehow expected, but plausibly not for all transcription factors. Many for instance are active only in phosphorylated form, meaning that the transcript can be constitutively expressed but the activity will depend on the phosphorylation status, and the correlation with targets should be reduced. In this way, one could derive a general model to relate the activity of a TF and its transcript's abundance, and in case the correlation is significant, to identify those transcription factors significantly deviating from the general behavior.

Data source

<https://www.nature.com/articles/s41598-022-12463-3>

<https://github.com/SBRG/precise-db/tree/master/data>

file to download: `log_tpm_norm.csv`

and for the regulatory network:

<https://regulondb.ccg.unam.mx/datasets>

5. DNA replication introduces copy number variations that are detectable in

RNA-seq, The extent of this signal depends on the growth rate, being visible when the division time ($=\ln 2/\text{growth_rate}$) is shorter than the time required to replicate the chromosome (speed of the DNA polymerase $\sim 1000\text{nt/s}$, length of one chromosome arm $\sim 2\text{E}+6$, time required $\sim 1/2\text{h}$, around 40min), which is possible only in mero-oligoploid species, like *E. coli* or *B. subtilis*. These copy number variations produce a clear pattern in gene expression also in single cell data (see for instance <https://www.biorxiv.org/content/10.1101/2022.10.22.513359v1>) but here you can use the gene expression compendium called log_tpm.csv in archive

data_coli_growth_rate.zip; the other csv file contains information on the samples, in particular on the growth rate (you will understand why at the end).

The objective of this project is to check if indeed the effect of replication is visible at the transcriptome level. The easiest way is to translate gene coordinates in radians (on the basis of the circularity of the genome), such that then one can build models of the type $y = \alpha + \beta \cos(\text{rad}X) + \gamma \sin(\text{rad}X)$ for each condition in the gene expression compendium, where Y is the gene expression level of a gene and X its coordinate (therefore both X and Y are vectors). To highlight this signal and remove some of the noise inherent in this kind of data, you can group genes located in a certain genome window (e.g. 100kb), calculate a mean gene expression level for the windows (in case, in log such that then the mean corresponds to the weighted mean in the original scale) and then build a model of the type above for each condition.

When the pattern is present, one should get beta and or gamma parameters (referring to the same notation used for the model above) that are significantly different from zero. How are these significant values distributed? What happens if the models account only for the cos or only for the sin? What the so estimated parameter means? Do you think one can derive a quantitative relationship of the parameter from the model and the growth rate? Implement some of the analysis above, and discuss what you get.

The interesting thing, if you get at least some conditions with a significant fitting when, comes out from the following theoretical treatment:

Model of copy number based on cell cycle “phase” duration (C=DNA synthesis phase, D predivisional phase), doubling time ($\tau = \log(2)/\text{growthrate}$) and position of a locus is (if interested, see works by Helmstetter and Cooper, 1968 and later):

$$n_a = 2^{((p_a C + D)/\tau)}$$

where

n_a is the copy number of locus a positioned at a distance p_a from the terminus of the chromosome (fractional distance such that the origin has a coordinate 1 and the terminus a coordinate 0). You can immediately see that $\log_2 \frac{n_{ori}}{n_{ter}} = \frac{C}{\tau} = \frac{C\mu}{\ln 2}$ therefore

we can get dynamic information about the speed of growth of a bacterium by calculating this ratio e.g. from DNA sequencing. In your case, since you deal with gene expression, and single cell data (if you use those), the situation is slightly different: the log ratio of the maximum and minimum of the fitted function will be something having to do with the growth rate as above, and you can evaluate this by comparing it with the growth rate of the different samples that you find in the metadata file within the same archive where you got the compendium of gene expression (called sample_table.csv – be careful, there are empty fields, so do not merge consecutive delimiters, and delimiter is comma). On this, samples where the growth rate is set to 1 should not be used as they use this number to indicate samples for which they don't have growth rate measurements! (stupid enough right?). If you do the project in 2 or 3 people, you may use those samples to try to predict their growth rate on the basis of what you get for the samples that are instead growth-rate informed.

Data file: data_coli_growth_rate.zip in the myariel folder where you downloaded this file.

6. Prediction of the number of 16s rRNA genes based on available genomes.

The data I am providing come from

https://rncdb.umms.med.umich.edu/genomes/?order_by=-dsorgname and can be used to develop your own 16S rDNA gene copy number predictor. The files in the 16S.zip here in this folder are a fasta file containing 16S rDNA sequences, and a table containing the number of copies of the 16S in the genome for a lot of species (I removed a few columns, therefore if you want to re-download the file from the database consider this). For each species you have min, max, mean mode and median of the number of copies, you can use the mean for your predictions. The idea is to predict the number of copies of genomes iteratively making as if you did not know the actual number of 16S of that species. A plausible set of steps for achieving this could be: a) randomly select one genome to be predicted – I will call it the target organism; b) identify let's say 20-30 genomes that are the most closely related to the target organism (for instance, to do this you can use the average identity percentage of their sequences– remember that you can have multiple sequences per genome in the fasta file) c) align the selected sequences comprised ONE from the target organism, use it to d) build a phylogenetic tree and e) use it together with the information on the number of copies in all the non-target organisms, to predict the number of copies in the target organism. f) store the predicted and true value for calculating error rate at the end.

If one repeats the procedure sketched above for a large enough sample of target organisms, he/she could get an idea of the performances of the predictor,

For step (c) you can call a system software like mafft or muscle from your R or python script, or use some basic aligner implemented in R

<https://cran.r-project.org/web/packages/orthGS/vignettes/Performing-Sequence-Alignment-in-R.html>

To build the phylogenetic tree from the alignment, you can use simple reconstruction methods like those from package adegenet:

<https://cran.r-project.org/web/packages/adegenet/index.html>

and for prediction exploiting the tree, check the following:

<https://www.rdocumentation.org/packages/picante/versions/1.8.2/topics/phyEstimate>

<http://blog.phytools.org/2016/10/estimation-of-missing-tip-states-under.html>

and

<https://blog.phytools.org/2019/03/phylogenetic-imputation-with.html>

Using functions from Phytools, one can also get an idea of how fast the number of 16s rRNA genes changes in time, by estimating for instance the rate matrix (but this is another story).

7. Relationship of the percentage of identity in the full 16s gene with respect to identity calculated on parts of the gene (e.g. at least V3-V4, V8, V9). In this way you can check if using the same threshold for the full gene or only part of it (V3-V4), gives the same results or not, and you can estimate a correction when using regions to achieve the same threshold wrt the full gene.

You can use the sequences in 16S.zip. A possible protocol is: 1. align a subset of these sequences using mafft in fast mode (or some other fast aligner). 2. Use functions from package seqinr to load the alignment and calculate all pairwise distances. 3. Find the coordinates of regions V3 and V4, cut the corresponding part from the alignment (this can be also done once the multialignment has been loaded in R), 4. calculate pairwise identities as before, but now only for the regions. 5. Build a regression model of the identities obtained in step (2) and (4) focusing only on identities above let's say 85% (you can test several thresholds to see if the parameters of the regression change). Discuss the results. If the x and y were the same thing (as usually is assumed by people doing metagenomics), then one should get a very high R-square (which I bet it is the case because we know the two things are surely correlated), and a parameter beta ~ 1 . Is the estimate obtained significantly different from 1? For this you can exploit the standard error of the estimate as provided by function lm in R. What about plotting the histogram of $\log_2 \text{id_full/id_V3-4}$? if the beta parameter is significantly different from one, how we can interpret the result? What this means in terms of previous data analysis? What would be a more precise threshold to be applied to the V3-4 region only to be congruent with a 97% threshold when comparing the full length 16s?

If you really want to do things the right way, maybe you can account for the non-independence of observations (pairs full – partial 16S rRNA) by first calculating a phylogenetic tree from the subset of 16S genes you are using, and then performing phylogenetic regression instead of the normal one. This can be accomplished in three ways, (1) using the tree as input in one of the possible function performing regression on a tree (like phylolm), (2) calculating the variance-covariance matrix from the tree and then using it as input to gls, (3) calculate phylogenetic contrasts (e.g. using pic from package ape) and then use them in lm for linear regression.

8. Relationship fitness of a transcription factor with the fitness of its targets.

We explored a little bit this issue in class, by focusing on transcription factors regulating a single gene. This on one side enabled us to only consider TF regulating the same number of genes, therefore we could build a single model e.g. linear $y \sim a + b \cdot x$ where y was the fitness of the TF and x of the target, in many different conditions (in class we were focusing on a single condition). You can start with this, but as in class, you will see that the downside of selecting TFs this way is that you reduce a lot the number of observations, especially because we need to consider differently positive and negative regulators. Considering TFs with more targets introduce a lot more problems in the analysis, but can be done. Here I will provide a few suggestions of increasing difficulty. A) you can test different ways of combining the fitness of multiple targets to reduce this analysis to the one target case; for instance you could calculate mean, max and product of the fitness values for all targets, and then check which is better in predicting the fitness of the regulator in regression models, and then “classify” the different regulators on the basis of which predictor is better (there would be TF for which the best predictor is the mean, the max...). You will see that in this way only a few Tfs will get a significant model. B) build regression models separately for each regulator like this:

$y \sim a + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$, where y are the fitness values of a regulator across the many conditions you have in the data, and xs for the targets. One could also try to use interaction terms in the model but when the number of targets increases, you will

get a long list of predictors, and in that case it would be better to use some regression technique that enables you to reduce the number of predictors (e.g. stepwise regression or LASSO regression). In the end you would get one model per TFs, and you need to summarize the results to extrapolate your conclusions. One basic way is to check how many times indeed the targets are assigned a significant regression coefficient for the TFs (this would give you an idea about our hypothesis being correct), and you can also study the distribution of the Rsquare for the significant models. Remember, if you build a regression model with 10 predictors, and then only one of them is significant, you should rebuild the model with only that predictor, if you want to get an estimate of the R-square; otherwise the R-square could be even very high because of the presence of many parameters which are however not significantly different from 0 (and therefore, they are 0 in practice, from the statistical point of view). Moreover, if you try with interaction terms, another interesting question you may want to answer is how many times the interaction terms are significant, which could indicate epistasis. This project, is the less well defined of the list, it has a lot of open points that are not easily defined; therefore, you have to invent some approach to do something new, and it will not be very easy because the situation is quite complicated. But, I don't ask to obtain results publication ready, but that you try to implement your own analysis making choices that are appropriate for the problem at hand and then discuss your results also considering the limits.